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Nama : Hijriyani Nur Riska

"Perbaikan"

No.
Date:

Latihan Soal - Soal Resistor !

1. Berapakah nilai tahanan untuk resistor berikut :

a) Kuning, Ungu, Emas, Perak

$R = 47 \times 10^{-1} \pm 10\%$

$= 4{,}7\ \Omega \pm 10\%$

b) Orange, Putih, Kuning, Emas

$R = 39 \times 10^{4} \pm 5\%$

$= 390.000\ \Omega \pm 5\%$

$= 390\ k\Omega \pm 5\%$

c) Coklat, Hitam, Merah, Emas

$R = 10 \times 10^{2} \pm 5\%$

$= 1000\ \Omega \pm 5\%$

$= 1k\Omega \pm 5\%$

d) Coklat, Hijau, Hijau, Tidak Berwarna

$R = 15 \times 10^{5} \pm 20\%$

$= 1.500.000\ \Omega \pm 20\%$

$= 1{,}5\ M\Omega \pm 20\%$

e) Hijau, Biru, Hitam, Merah, Coklat

$R = 560 \times 10^{2} \pm 1\%$

$= 56000\ \Omega \pm 1\%$

$= 56k\Omega \pm 1\%$

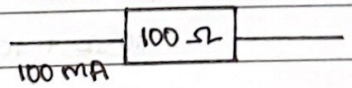
f) Coklat, Abu-abu, Hitam, Hitam, Coklat

$R = 180 \times 10^{0} \pm 1\%$

$= 180\ \Omega \pm 1\%$

2. Hitunglah daya yang didisipasikan pada masing-masing tahanan dengan kondisi tahanan di bawah :

a)



100 mA = 0,1 A

$P = I^2 \times R$

$= (0{,}1)^2 \times 100$

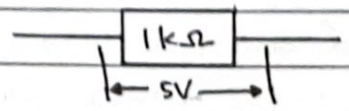
$= 0{,}01 \times 100$

$= 1$ Watt

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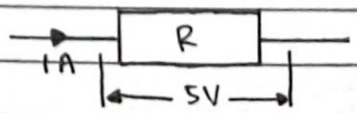
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b)



$P = \frac{V^2}{R} = \frac{(5)^2}{1000} = \frac{25}{1000} = \frac{1}{40} = 0{,}025$ Watt

c)



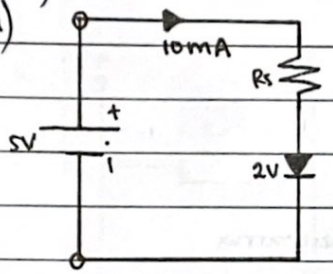
$P = V \times I$

$= 5 \times 1$

$= 5$ Watt

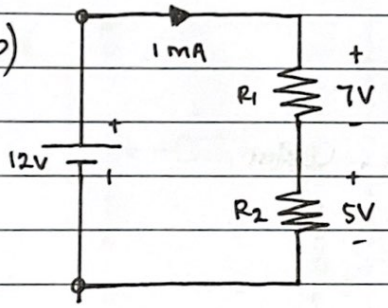
3. Tentukan nilai resistor pada rangkaian di bawah ini yang sesuai dengan nilai standar resistor E12.

a)



$R = \frac{V}{I} = \frac{5}{10 \cdot 10^{-3}} = \frac{5}{10^{-2}} = 5 \times 10^2 = 500\ \Omega$

b)



$R_1 = \frac{V}{I} = \frac{7}{1 \cdot 10^{-3}} = 7 \cdot 10^3 = 7000\ \Omega = 7k\Omega$

$R_2 = \frac{V}{I} = \frac{5}{1 \cdot 10^{-3}} = 5 \cdot 10^3 = 5000\ \Omega = 5k\Omega$

Jadi, $R_{total} = R_1 + R_2$

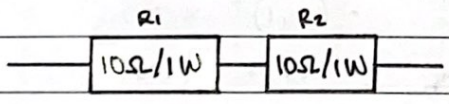
$= 7K\Omega + 5K\Omega$

$= 12K\Omega$

4. Hitung besar tahanan total dan daya total bila resistor 10 Ω / 1 W disusun secara :

$W_{total} = 1 + 1 = 2W$

a) Seri



$R_{total} = R_1 + R_2$

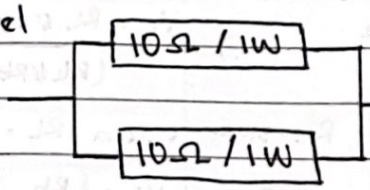
$= 10\ \Omega + 10\ \Omega$

$= 20\ \Omega\ /\ 2W$

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b) Pararel

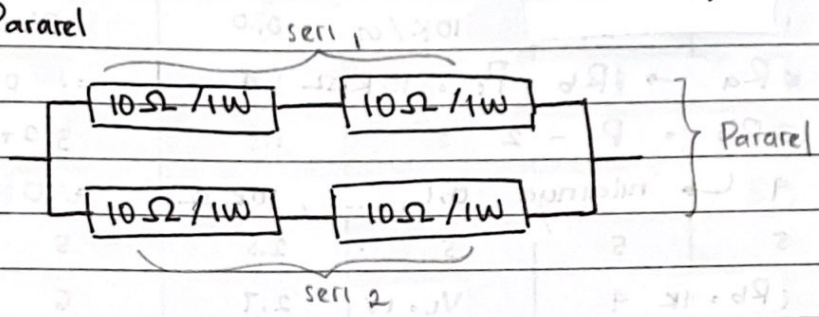


$\frac{1}{R_{total}} = \frac{1}{10} + \frac{1}{10} = \frac{2}{10} = \frac{1}{5} = 5\Omega$

$W_{total} = 1 + 1 = 2W$

∴ Jawabannya adalah $5\Omega / 2W$

c) Seri - Pararel



$R_{S1} = 10 + 10 = 20\Omega$, $W_1 = 1 + 1 = 2W$

$R_{S2} = 10 + 10 = 20\Omega$ $W_2 = 1 + 1 = 2W +$

$4W$

$R_p = \frac{1}{R_{S1}} + \frac{1}{R_{S2}}$

$= \frac{1}{20} + \frac{1}{20}$

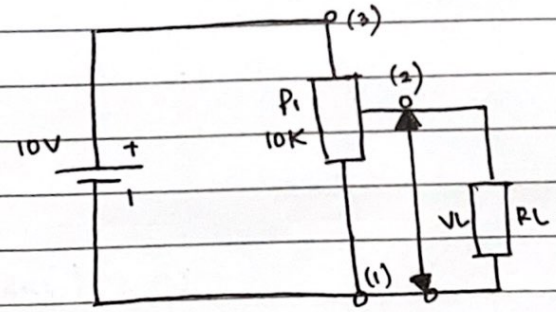
$= \frac{2}{20}$

$= \frac{1}{10}$

$= 10\Omega / 4W$

5. Gambar Gravik $VL = f(P_{1-2})$ untuk kondisi :

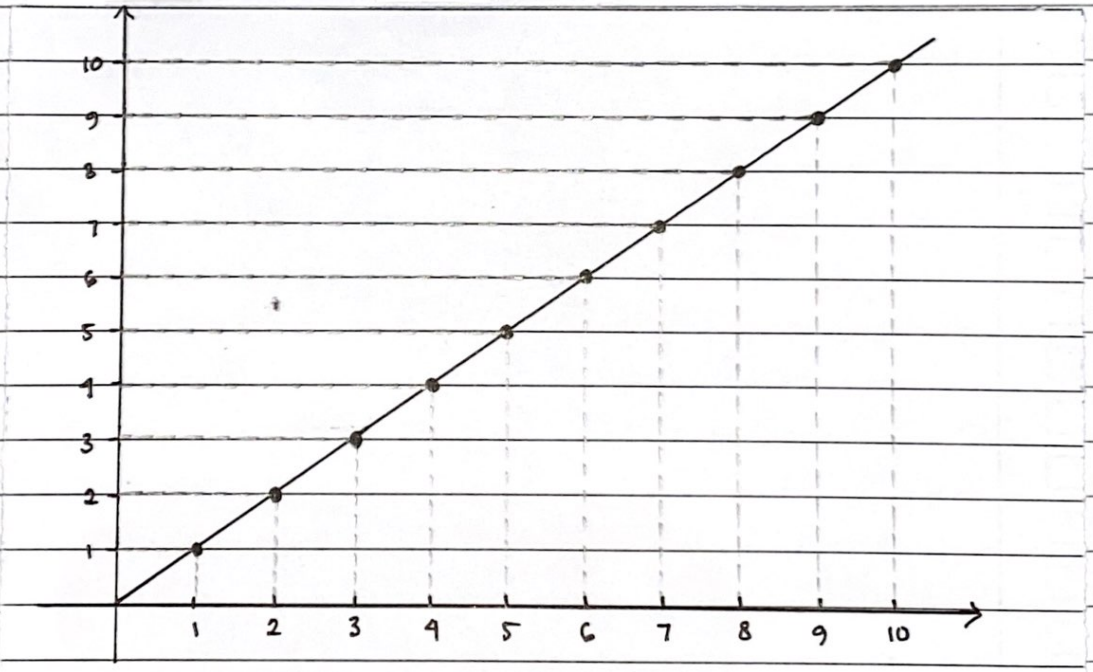
a) $R_1 = \infty$ (tak hingga)



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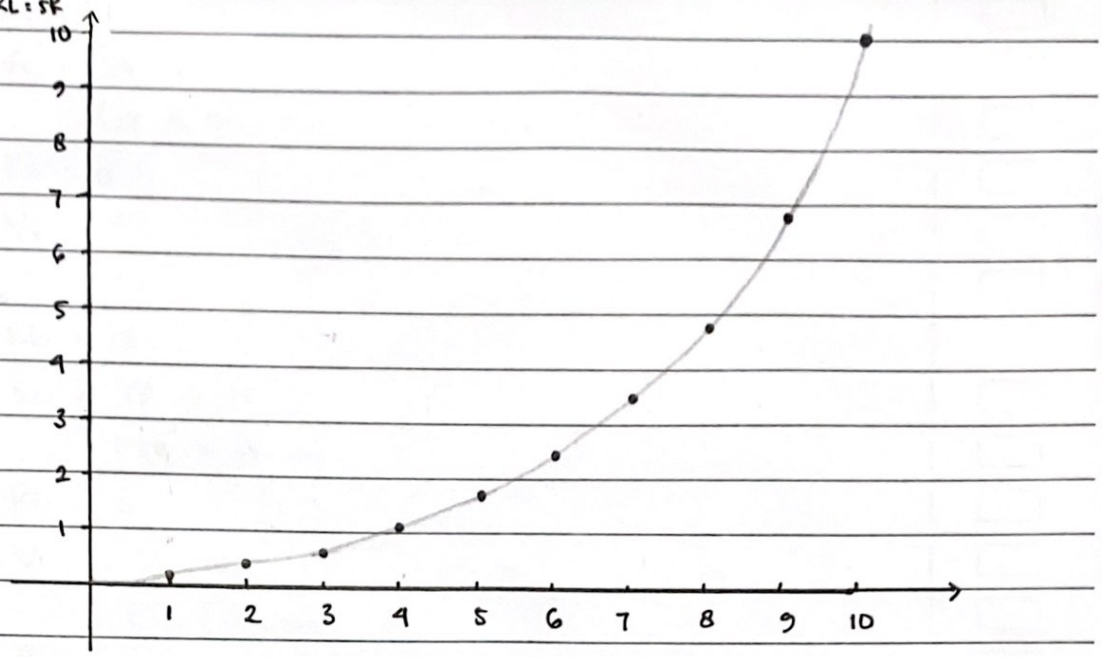
Nilai Potensiometer		$R_p = P_{12} \,//\, RL\ (K\Omega)$		$VL = \frac{R_p}{R_p + P_{25}} \times 10\,V$	
$P_{12}\ (K\Omega)$	$P_{25}\ (K\Omega)$	RL ∞	RL = 5KΩ	VL dgn RL ∞	VL dg RL 5KΩ
0	10	0	0,0	0	0,0
1	9	1	0,8	1	0,1
2	8	2	1,4	2	0,3
3	7	3	1,9	3	0,6
4	6	4	2,2	4	1,1
5	5	5	2,5	5	1,7
6	4	6	2,7	6	2,4
7	3	7	2,9	7	3,5
8	2	8	3,1	8	4,8
9	1	9	3,2	9	6,9
10	0	10	3,3	10	10,9



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b) RL = 5K



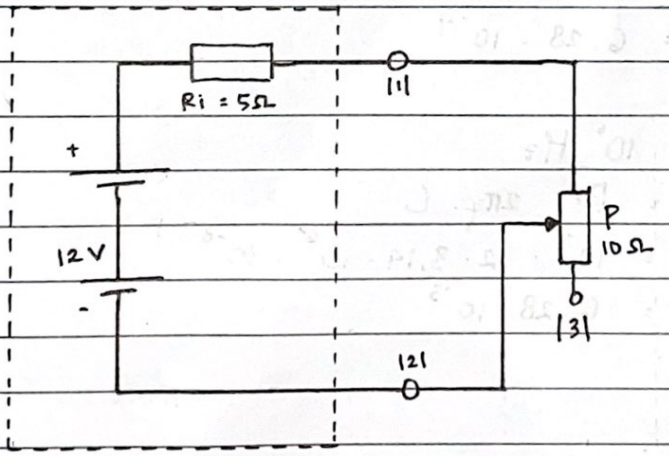
6. Nilai Resistor variabel berubah dari 0 - 10 Ω [P1-2]

Gambarkan pada diagram yang sama :

a) Grafik daya yang didisipasikan pada Potensiometer P(pot) = f [P1-2]

b) Grafik daya yang didisipasikan pada tahanan dalam baterai
P(Ri) = f (f [P1-2])

c) Dari hasil grafik yang dibuat pada soal a) dan b), pada kondisi mana terjadi transfer daya maksimum pada beban potensiometer?

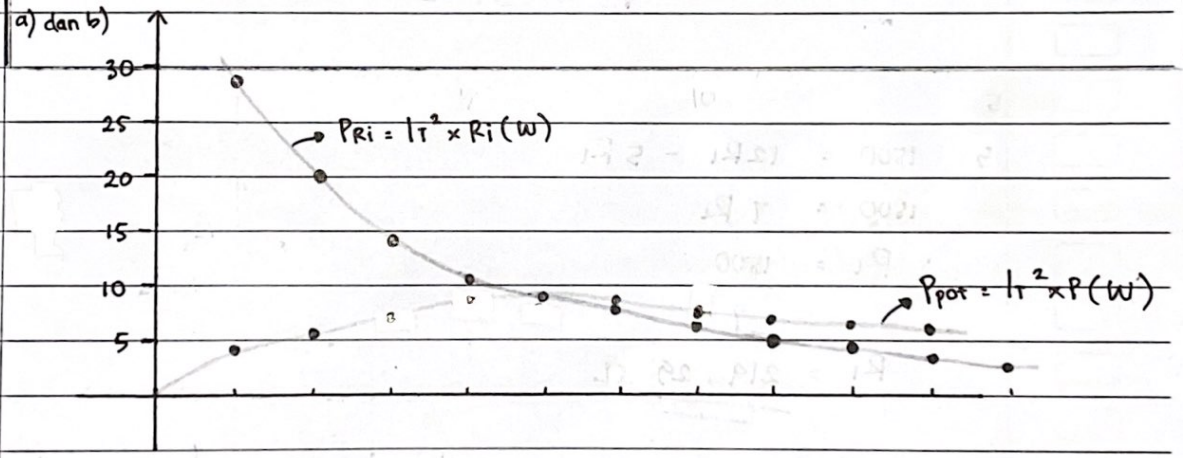


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P(Ω)	Ri(Ω)	$R_T = P + R_i$ (Ω)	V_{IN} (V)	$I = \frac{V_{IN}}{R_T}$ (A)	$P_{Ri} = I_T^2 \times R_i$ (W)	$P_{pot} = I_T^2 \times P$ (W)
0	5	5	12	2,4	28,8	0,0
1	5	6	12	2,0	20,0	4,0
2	5	7	12	1,7	14,7	5,9
3	5	8	12	1,5	11,3	6,8
4	5	9	12	1,3	8,9	7,1
5	5	10	12	1,2	7,2	7,2
6	5	11	12	1,1	6,0	7,1
7	5	12	12	1,0	5,0	7,0
8	5	13	12	0,9	4,3	6,8
9	5	14	12	0,9	3,7	6,6
10	5	15	12	0,8	3,2	6,4

a) dan b)



c) Transfer daya maksimum pada beban potensiometer terjadi pada saat P dan Ri bernilai 5Ω. Daya maksimum yang didapat yaitu sebesar 7,2 Watt.

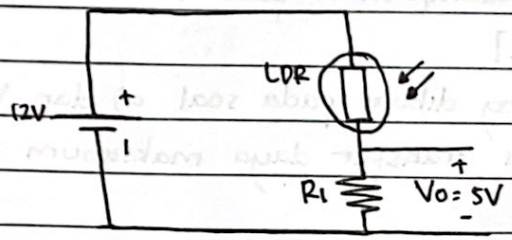
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7. Tentukan besar nilai R_1 supaya $V_o = 5V$ dalam kondisi terang, bila spesifikasi LDR mempunyai :

Tahanan Gelap (Dark Resistance) = 2 MΩ

Tahanan Terang (Light Resistance) = 300 Ω



Jawaban :

$$V_o = V_s \times \frac{R_1}{R_1 + R_{LDR}}$$

$$5 = 12 \times \frac{R_1}{R_1 + 300}$$

$$5(R_1 + 300) = 12R_1$$

$$5R_1 + 1500 = 12R_1$$

$$1500 = 12R_1 - 5R_1$$

$$1500 = 7R_1$$

$$R_1 = \frac{1500}{7}$$

$$R_1 = 219,29\ \Omega$$